

ME350 FA 2025 Section 002

Lab #1

Monday 12:30 PM - 2:30 PM

Haoran Shi

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Office Hours: 3:00 PM - 5:00 PM Monday (Zoom)



Haoran Shi

Background

- Fourth year Ph.D. student in ME in Manufacturing
- Working on simulation of the metal additive manufacturing process
- ME M.S. University of Michigan (2020-2022)
- ME B.S. Rose-Hulman Institute of Technology (2016-2020)

Hobbies

- Cycling
- Flying airplanes



Agenda

- Reminders
- Project Description Overview
- Project Schedule
- Self-Paced Lab

Reminders

- Submit team preferences [\[FORM\]](#)
- Due Wednesday (9/3) by 11:59 pm
- Team expectation agreement due with Lab Assignment 2

- Homework 1 Due September 9th at 1:30 pm (start of lecture)
 - Submit to Gradescope
 - Office hours: [\[LINK\]](#)



Reminders

- Please obtain the following required course materials as soon as possible:
 - Safety glasses (with a Z87 rating)
- To bring to class for the next few lectures (sold as a package at campus bookstores):
 - 2 plastic triangles (with any angles)
 - A ruler
 - A protractor
 - A compass (for drawing arcs of circles)
- Four required training modules link:
 - [\[LINK\]](#)
 - Scroll to the bottom to ensure you have all four!



Slack

- We will be using Slack for all communication within the class this semester

How to join:

- Go to um-fa25-ME350.slack.com
- Sign in using your M-login

You will receive an invite to your lab's private channel shortly after

Calendar

- Scroll down on Canvas homepage to find the full calendar

Week		Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
			Aug 25	Aug 26	Aug 27	Aug 28	Aug 29	Aug 30
1	Lecture			Lecture 1 - Course Overview and Logistics (ALL)		Lecture 2 - Mechanisms: Kinematic Diagrams and Mobility (DB)		
	Homework					HW1 Assigned		
	Lab		Lab 1 - Project Overview, SolidWorks Graphical Linkage Synthesis and Sketch Relations Tutorial (CAEN lab)			Lab 1 - Project Overview, SolidWorks Graphical Linkage Synthesis and Sketch Relations Tutorial (CAEN lab)	Lab 1 - Project Overview, SolidWorks Graphical Linkage Synthesis and Sketch Relations Tutorial (CAEN lab)	
	Project Timeline		Classes Begin - FA25					
September		Aug 31	Sep 1	Sep 2	Sep 3	Sep 4	Sep 5	Sep 6
2	Lecture		Labor Day	Lecture 3 - Mechanisms: Kinematic Inversions, Instant Centers (DB)		Lecture 4 - Mechanisms: Mechanical Advantage and Graphical Synthesis (DB)		
	Homework							
	Lab				Team Preferences Form is Due	Lab 2 - ADAMS Tutorial (CAEN Lab) Lab Assignment #1 Due - Sketch Blocks	Lab 2 - ADAMS Tutorial (CAEN Lab) Lab Assignment #1 Due - Sketch Blocks	
	Project Timeline							

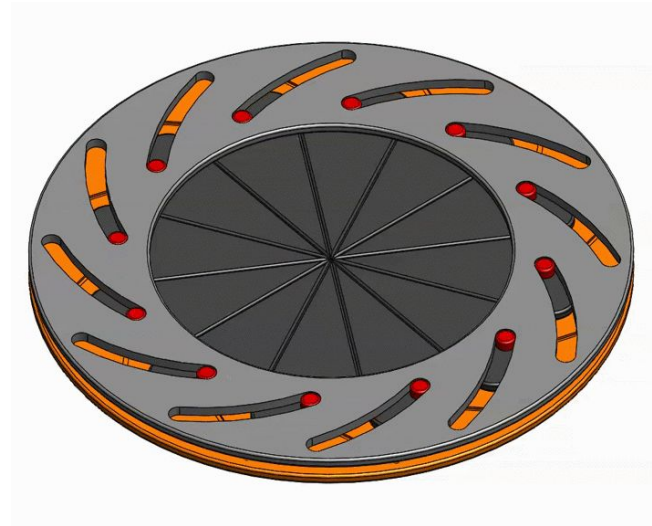
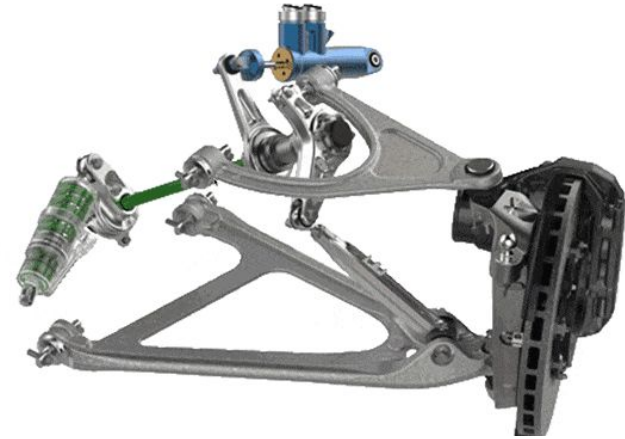
Project Overview

Project

- Overarching Goal:
 - Design/build/test powered mechanism to sense activated targets and respond by shining light on corresponding photosensors to deactivate.
- Contest Goal:
 - Deactivate targets as quickly as possible
 - [\[ME350 Video\]](#)
 - **Note:** we won't be doing distributed manufacturing this semester!
- See full project description on Canvas!!
 - Canvas > MECHENG 350 001 FA 2025 > Files > Project



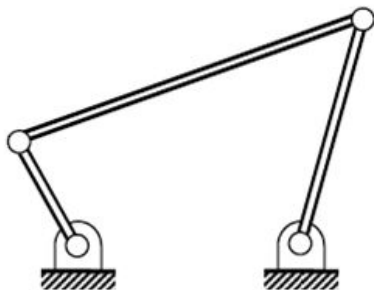
Real-World Linkage Examples



Modules

Motion
Generation

SolidWorks
Adams



Transmission

Motors
Gears
Pulleys
Belts

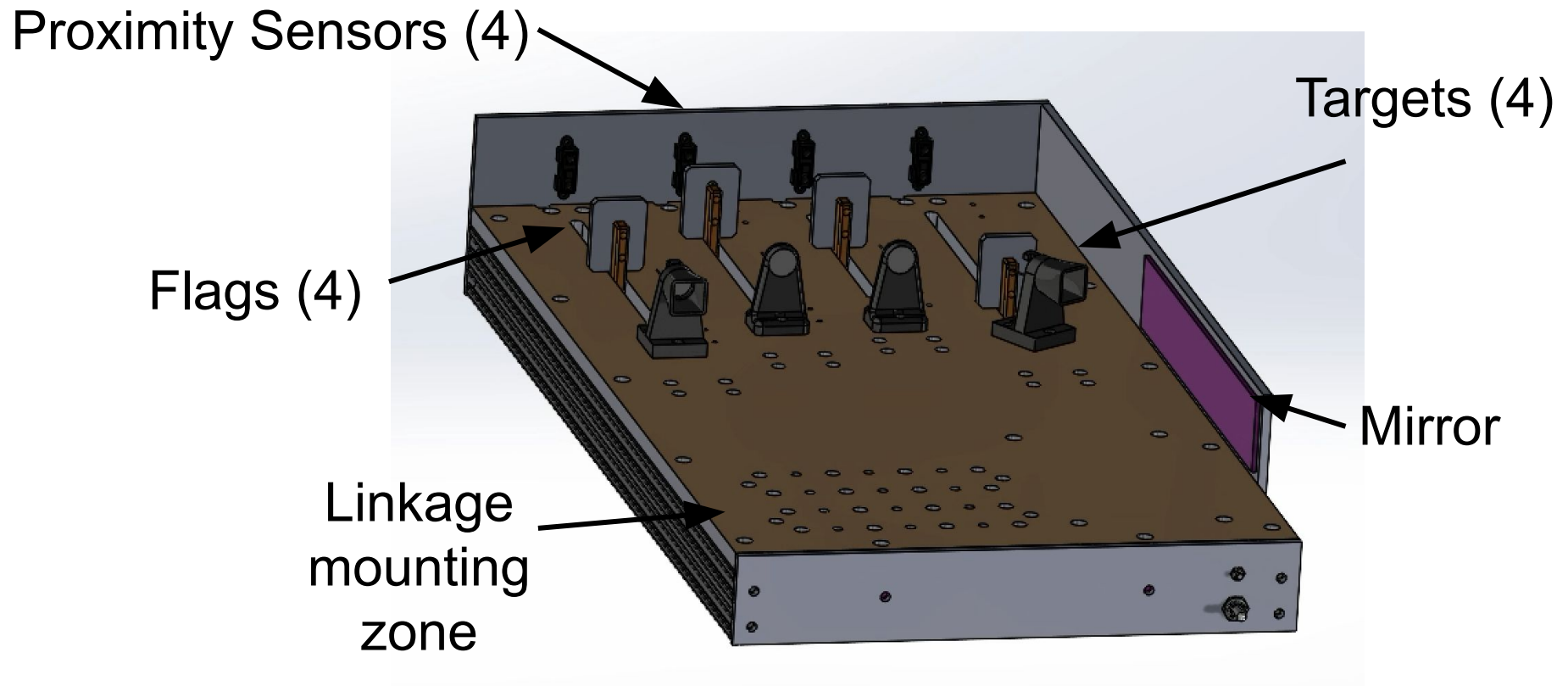


Motor Control

Arduino
Sensors
Switches

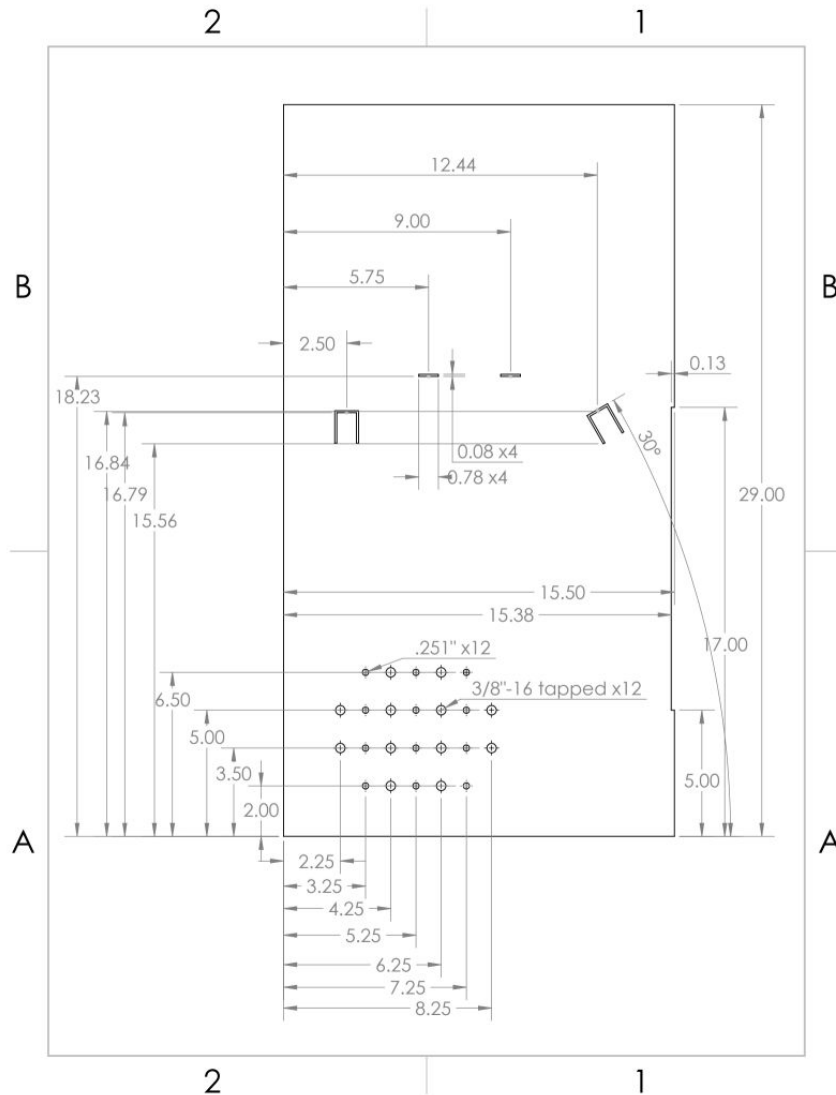


Playing Field



Take your own measurements!
(Playing fields in X50 assembly room)

Playing Field Dimensions



Note: use the provided drawing in section specific Canvas files for a full view, also linked [here](#).

Project Specifications

- Mechanism Evaluation

Specification	Description	Target	% of Grade	Method of Measurement
Transmission Angle Deviation	<i>Transmission Angle</i> - 90° (average of all five targets)	0°	20	Machinist's Protractor
Weight		Minimize	30	Scale
Performance Score	Based on number of targets dropped and amount of light on targets	Maximize	50	Playing Field

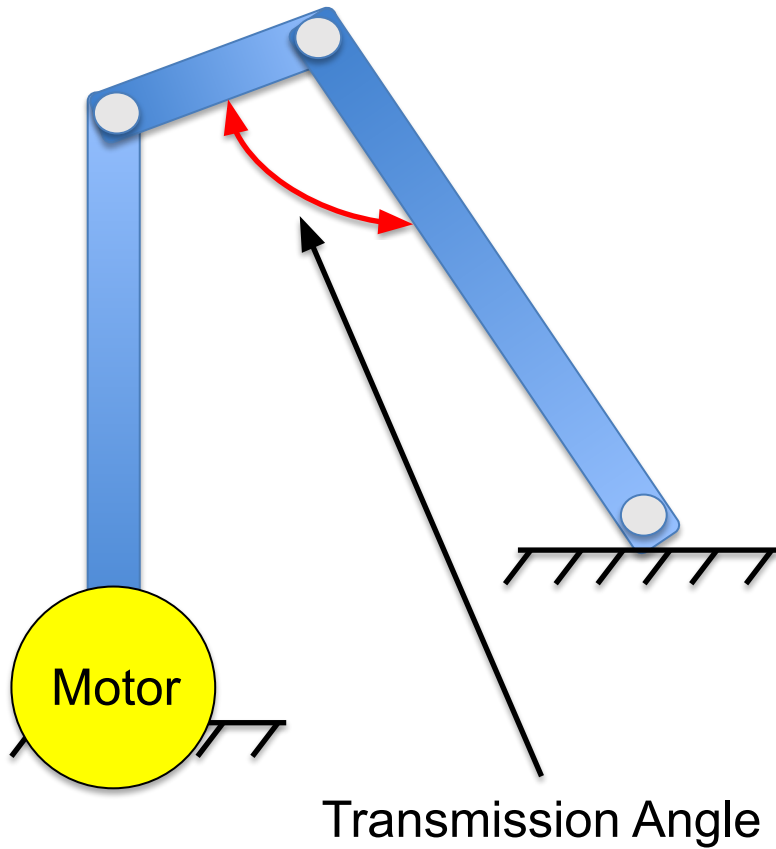
- Performance Scoring During Game

Action	Measurement	Scoring
Retreat Target- R1	Number of targets successfully sent back in R1	Measurement x 1
Retreat Target- R2	Number of targets successfully sent back in R2	Measurement x 1
Retreat Target- R3	Number of targets successfully sent back in R3	Measurement x 1

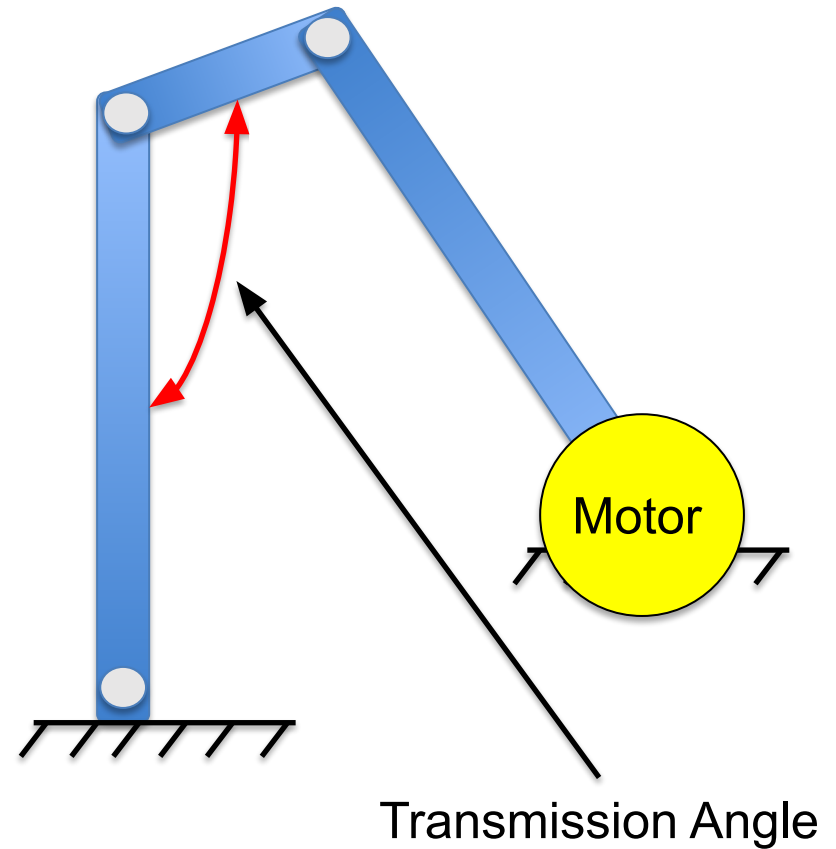


Transmission Angles

Option #1



Option #2



Why are Transmission Angles so important?

- In this project, we measure the absolute transmission angle **deviation from 90°**
- As the transmission angle itself approaches 90° , generally the mechanical advantage of the linkage improves
- Mechanical advantage is a concept that will be covered in lecture: it is the ratio of output force (where end effector is) to input force (where the actuator acts)
- Also, if the transmission angle approaches 180° , the mechanism reaches a toggle point where the mechanism either locks or behaves erratically

Course Grading

50% - Project	30% - Final Report	
	10% - DR1	
	15% - DR2	
	25% - Test Results	5% - Transmission Angle Deviation 7.5% - Weight 12.5% - Performance Score
	20% Craftsmanship	10% Manufacturing Team 10% Design Team
15% - Midterm Exam		
20% - Final Exam		
15% - Homework + In-Class Activities		

Design Reviews

- Gate Review
 - Opportunity for feedback on mechanism
 - Review design prior to fabrication
 - Informal, outside of lab time
 - GSI will set up times
 - Tool to document progress



Questions?

Today

- Run through [Lab 1 Tutorial](#)
 - NOTE: please work through this lab with careful design consideration
 - Each mechanism design will be a candidate for your team's final design!
So make it good!

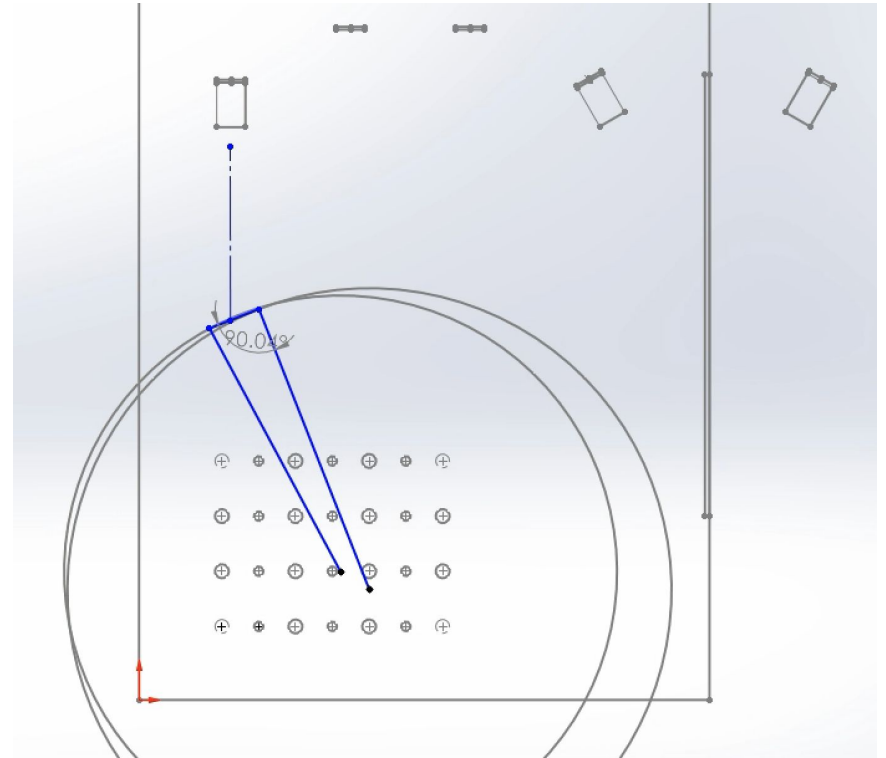
Due next lab:

1. [Submit team preferences to GSI](#)
2. Fill out the [Lab Assignment 1 Worksheet](#)
3. Join the [ME350 Slack channel](#)



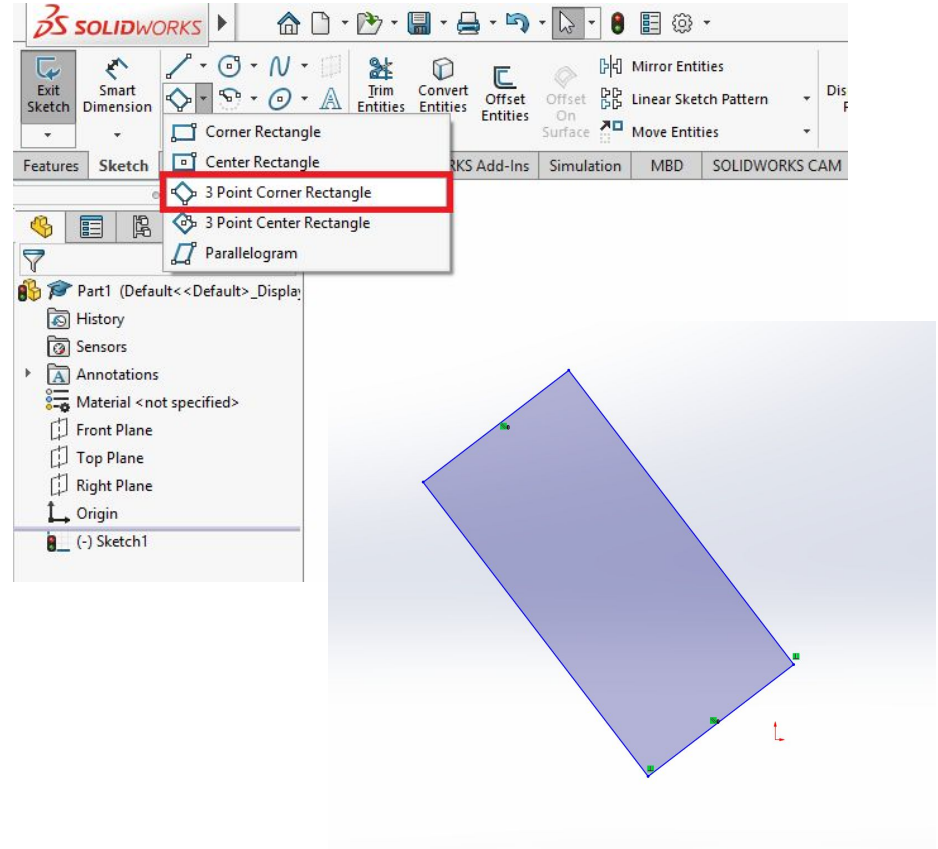
Designing Your Linkage

- Total weight
 - Minimize, but keep structural integrity and waterjet time in mind!
- Link inertia
 - Minimize
- Trans. angle deviation
 - As close to 0 as possible
- Angle of light on target
 - Think about the best angle



Pro Tip

- Use 3 Corner rectangle tool for making angled rectangles
 - There are different ways to make the angled rectangle, ask me if you have questions



Solidworks sketch relations/constraints

- **In Lab 1, be careful about sketch relations!**
 - When sketching, unwanted relations may be assigned! Know how to recognize these.
 - Be intentional about using the **fix** relation.
 - This 7-minute [video](#) is a great resource for those who may be more unfamiliar with Solidworks.
- **Another Pro Tip**
 - Final sketch mechanism needs to be **under defined** to be able to move it around
 - **Blue is under defined**
 - **Black is fully defined**